Hootan Seifi

AI Engineer · LLM & Agentic Systems · Python, FastAPI, LangGraph, RAG

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Live portfolio: hootanseifi.pro | **GitHub:** github.com/hootanseifi | **LinkedIn:** linkedin.com/in/hootan-seifi-42b180124
Open to on-site, remote, hybrid and project-based roles

SUMMARY

AI Engineer who builds and ships LLM systems end to end: agents, RAG, evaluation, guardrails and deployment. My live portfolio, **hootanseifi.pro**, runs a resume-grounded digital twin and a team of seven LangGraph trading agents in production for about **$3/month** in AI cost, backed by real-model evals (72/72, 20/20), **225 backend tests** and CI/CD with automatic rollback. M.Sc. in Electrical Engineering (Artificial Intelligence), plus earlier R&D, app review and financial-analysis roles, where I built a Python trading bot.

SKILLS

LLM & Agents: LangGraph, LangChain, OpenAI Agents SDK, CrewAI, MCP, tool calling, structured outputs, prompt engineering, LLM evaluation, guardrails, QLoRA / PEFT (Hugging Face TRL)

RAG & Retrieval: PostgreSQL + pgvector, hybrid search (vector + BM25, reciprocal rank fusion), embeddings, chunking, reranking, Chroma, Recall@k / MRR

Backend & DevOps: Python, FastAPI, SQLAlchemy, Alembic, REST, Server-Sent Events, Docker Compose, Caddy, Linux, GitHub Actions CI/CD, pytest, Playwright

ML & Data: PyTorch, scikit-learn, XGBoost, Pandas, SQL, Spark, Kafka, Airflow

Cloud & Frontend: AWS (Bedrock, SageMaker, Lambda), Terraform, Modal, Vercel, Next.js, TypeScript

EXPERIENCE

AI Engineer (Independent Projects) — Self-employed Aug 2026 – Present · Tehran

Design, build, test and deploy LLM and agentic applications end to end. All code is public on GitHub.

hootanseifi.pro — Live AI portfolio: digital twin, job-fit scorer, AI trading team hootanseifi.pro

- Built a bilingual (EN/FA) digital twin that answers only from my documents and cites its sources; added BM25 + pgvector hybrid search with reciprocal rank fusion after measuring weak Persian retrieval. Passes **72/72** eval questions, including refusals and prompt injections.
- Built a seven-agent LangGraph meeting that paper-trades Bitcoin at live prices. Risk rules (max 5x leverage, max 20% margin, 1–5% stop loss) are enforced in Python, not the prompt: **20/20** evaluated meetings with zero rule violations.
- Built a job-fit scorer whose score band is enforced in code from must-have coverage, so the model cannot flatter; temperature 0 for repeatable results.
- Kept AI cost at **~$3/month** with per-call daily and monthly budget caps and a run-once-per-day design: the first visitor triggers the meeting, everyone else watches the replay.
- Deployed 5 Docker services (Next.js, FastAPI, worker, PostgreSQL, Caddy) on a hardened VPS with nightly backups. GitHub Actions runs linters, **225 backend + 28 component + 15 end-to-end** tests and a secret scan, and rolls back a failed deploy automatically (tested with a deliberately broken release).

FastAPI · LangGraph · PostgreSQL + pgvector · SQLAlchemy · Next.js / TypeScript · Docker · Caddy · GitHub Actions

Astra — Enterprise AI agent platform github.com/hootanseifi/astra-ai-platform

- LangGraph agent that decides per request whether to use RAG, tools, both or neither, with a recovery node, planning, a PostgreSQL checkpointer and human-in-the-loop approval for sensitive tools.
- Agentic RAG on pgvector (HNSW index, reranking, permission-aware retrieval); reranking raised Recall@1 from **0.67 to 1.00** on the evaluation set.
- Security and operations: RBAC tool executor with allow-list and schema validation, JWT auth, prompt-injection detection (English and Persian), secret redaction, rate limiting, Prometheus metrics, token and cost tracking, and an MCP server.
- **407 tests, 93% coverage**, and an evaluation gate in CI (Recall@k, MRR, route accuracy, groundedness).

Price Estimate — ML price prediction and deal-hunting agents github.com/hootanseifi/price-estimate

- Re-implemented a course capstone (Ed Donner, LLM Engineering) as a tested Python package: a ~800k-product Amazon dataset, models from baselines to a deep residual network, a QLoRA fine-tuning pipeline for Llama 3.2 3B (Hugging Face TRL, served on Modal) and frontier-LLM RAG over Chroma.
- Replaced hand-picked ensemble weights with weights learned by non-negative least squares, with automatic re-weighting when a model is unavailable; fixed parser, upload and checkpoint bugs from the original notebooks.
- Benchmarked every model on the same test set with MAE ± 95% CI, RMSLE, R^2 and hit rate. Agents scan deal RSS feeds and send push alerts; served through a FastAPI */predict* endpoint and a Gradio UI, with 60+ offline tests in CI.

AI Medical Notes — SaaS for doctors github.com/hootanseifi/ai-medical-notes-saas

- Turns consultation notes into a visit summary, next steps and a patient email, streamed in real time over SSE. Next.js, FastAPI, OpenAI, Clerk authentication and subscriptions, deployed on Vercel.

Behengam — Integrated health platform (founder, in design) Interactive investor demo v2

- Connects 12 roles of Iran's health system around one shared patient record. Built the interactive demo and designed the product roadmap: HL7 v2 / FHIR ingestion through Kafka, LOINC / UCUM normalization, Django REST with role-based access, and an AI alert layer shown to doctors as suggestions, never decisions.

Digital Marketing & Content Manager — Eghtesad Gostar Behsaman May 2025 – Apr 2026

Senior Power Plant Specialist — Taban Energy Power Plant (Pasargad Bank) Mar 2025 – May 2025

Fundamental Analyst & Trader, Financial Markets — Novin Eghtesad Technologies Sep 2022 – Feb 2025

- Analyzed financial and crypto markets, evaluated blockchain projects for analytical reports, and developed a Python trading bot on market data.

Technical Specialist, App Review & Development — Cafe Bazaar Jan 2022 – May 2022

R&D Specialist, R&D Department — MAPNA Electric & Control Engineering (MECO) Apr 2020 – Mar 2021

EDUCATION

M.Sc. Electrical Engineering, Artificial Intelligence — Amirkabir University (Tafresh Branch) 2016 – 2018

B.Sc. Electrical Engineering, Electronics — Amirkabir University (Tafresh Branch) 2011 – 2015

Academic ML projects: fraud detection on Ethereum transactions with imbalanced data (XGBoost); tweet sentiment vs. Nvidia stock price (NLP, time series); traffic-sign classification with CNNs (GTSRB).

CERTIFICATIONS & LANGUAGES

LLM Engineering: Master AI, Large Language Models & Agents (Ed Donner, Udemy) · Machine Learning Specialization (Andrew Ng, DeepLearning.AI & Stanford) · Financial Trading in Python (DataCamp, 2025)

Languages: Persian (native), English (professional working proficiency)